

DATA QUALITY ASSESSMENT AND WORKING GROUP

Bristol Medical School

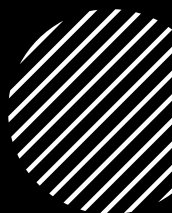
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Session Outline



What is data quality and why it matters

Dimensions Of Data Quality

Discussions

Working Groups - Using ANC Data

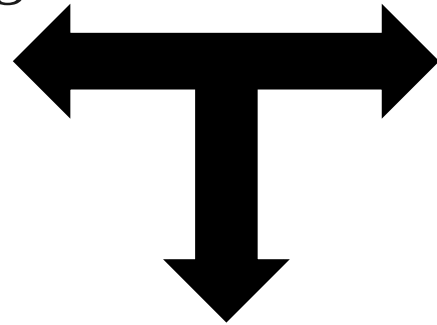
What is data quality?

- *Data quality is the degree to which a dataset fits the purpose it is being used for.*

It's contextual: Data 'good enough' for monthly reporting may be too rough for AI prediction. Quality is judged against intended use.

It has many faces:

Completeness, consistency, accuracy, timeliness, validity, uniqueness — a record can fail one without failing all.



It must be measured: 'The data looks fine' is not an answer. Quality is reported with numbers, per variable, per facility, per time period.

Why does it matter for our ANC project?



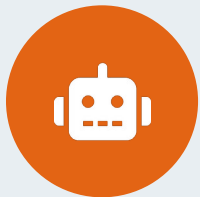
Care decisions rely on the record

If Hb is missing or wrong, anaemia goes unmanaged. If HIV status is mislabelled, the EMTCT cascade breaks.



Indicators reflect record quality, not just care

Reported ANC4+ coverage is only as good as the records that count visits.



AI cannot fix bad inputs

Models trained on incomplete or biased data make incomplete, biased predictions — and look confident doing it.



Publication demands rigour

Every reputable journal asks how quality was assessed. We need an answer beyond 'we cleaned the data'.

Six dimensions of data quality

Completeness:

- *Are required fields filled in?*
- *Are values missing?*

Consistency:

- *Do the values agree with each other?*
- *Are records similar across visits?*

Accuracy:

- *Do they reflect reality?*
- *Are values correct?*

Timeliness:

- *Is the right date in the right field?*
- *Are dates correct?*

Validity:

- *Do values conform to the expected format?*
- *Are values reasonable?*

Uniqueness :

- *Is each person counted exactly once?*
- *Are duplicates present?*

Completeness

THE QUESTION IT ANSWERS

Are the fields we need actually filled in?

HOW WE MEASURE IT

Per variable: percentage of records where the field is NOT missing. Per encounter: proportion of expected fields that are filled for one visit.

WHY IT MATTERS

A variable missing in 40% of records cannot be used as an AI predictor: and may signal a workflow problem in the clinic.

EXAMPLE FROM OUR DATA

Haemoglobin testing at ANC

```
anc %>%  
  summarise(hb_missing = mean(is.na(hb_levels_rslts)))  
# → 0.35 (35% of visits have no Hb result)
```

WHAT THIS TELLS US

More than one in three ANC visits in our dataset has no Hb recorded. Anaemia screening is a WHO standard: this is both a clinical and a data issue.

Consistency

THE QUESTION IT ANSWERS

Do the values in a record agree with each other?

HOW WE MEASURE IT

Logical rules: parity $\leq$ gravida, gestational age increases visit-to-visit, BP within plausible range. Count records that fail each rule.

WHY IT MATTERS

A record can be 100% complete and still be wrong. Consistency catches errors that completeness misses.

EXAMPLE FROM OUR DATA

Parity greater than gravida

```
anc %>%  
  filter(anc_parity > anc_gravida) %>%  
  count()  
# → 312 records (logically impossible)
```

WHAT THIS TELLS US

Parity cannot exceed gravida — you cannot have more births than pregnancies. These 312 records have a data-entry error and need review.

Accuracy

THE QUESTION IT ANSWERS

Do the recorded values reflect reality?

HOW WE MEASURE IT

Range checks against clinical plausibility. Cross-check against a known source where one exists.

WHY IT MATTERS

Outlier values bias means, distort distributions, and break models that assume normality. We catch them before they leak into analysis.

EXAMPLE FROM OUR DATA

Implausible blood pressure

```
anc %>%  
  filter(sbp > 220 | sbp < 60 |  
         dbp > 140 | dbp < 30) %>%  
  count() # → 47 records out of range
```

WHAT THIS TELLS US

Adult systolic BP outside 60–220 mmHg or diastolic outside 30–140 is almost certainly a data-entry error — not a real measurement.

Timeliness

THE QUESTION IT ANSWERS

Is the right date recorded in the right place?

HOW WE MEASURE IT

Check that dates are in the expected range, that 'event' and 'entry' dates differ as expected, and that no date is in the future.

WHY IT MATTERS

If we filter on the wrong date column, we lose data. Our own analysis nearly collapsed 22,000 records to a handful for exactly this reason.

EXAMPLE FROM OUR DATA

date_rcrd vs dent

```
# date_rcrd is when the row was ENTERED  
# dent      is when the VISIT happened  
# filtering on the wrong one removed 99% of data
```

WHAT THIS TELLS US

A real bug from our project. 'date_rcrd' is mostly 2025; 'dent' is the visit date in 2023-24. Naming and using the right date is part of data quality.

Validity

THE QUESTION IT ANSWERS

Do values conform to the expected format or code list?

HOW WE MEASURE IT

Check that categorical fields contain only allowed codes. Flag free-text entries in fields that should be coded.

WHY IT MATTERS

Free-text in checkbox fields is invisible to most analysis pipelines. Counts of 'yes' may exclude variants like 'YES', 'Yes ', or 'y'.

EXAMPLE FROM OUR DATA

Free-text mixed with codes

```
anc_diagnosis__2 values include:  
'Checked'  
'Unchecked'  
'ISS IN PREGNANCY ' ← free text!
```

WHAT THIS TELLS US

The diagnosis fields are designed as checkboxes but contain free-text in some rows. These are unusable for tabulation until standardised.

Uniqueness

THE QUESTION IT ANSWERS

Is each person, visit, or outcome counted exactly once?

HOW WE MEASURE IT

Find duplicate rows. Identify identifiers (e.g. client_no) used for multiple distinct people. Use linkage to deduplicate.

WHY IT MATTERS

Without uniqueness, every per-patient metric — ANC4+, retention, outcomes — is wrong. AI models will see 'patients' that don't exist.

EXAMPLE FROM OUR DATA

client_no = 189

```
anc %>% filter(client_no == 189) %>%  
  distinct(client_name)  
# → 7 distinct names, ~4 different women
```

WHAT THIS TELLS US

One client number was used by four different women had to build a probabilistic linkage to recover the real patients.

Three kinds of missingness

Within completeness, not all missingness is equal. The pattern determines how we should handle it.

M C A R	M A R	M N A R
Missing Completely At Random: The missingness is completely independent of both observed and unobserved data. EXAMPLE <i>Reagent ran out for a week, so Hb is missing across all visits in that period.</i>	Missing At Random: The missingness is systematically related to observed data but not to the missing data itself. EXAMPLE <i>Hb less often recorded in third trimester (depends on observed gestational age).</i>	Missing Not At Random: The missingness is directly related to the missing data itself. EXAMPLE <i>HIV status missing because the woman declined the test (the value drives the missingness).</i>

Count missing values AND test whether the missingness is patterned. Both matter.

Group Work

Instructions:

- In your small subgroups, identify the five most common problems that you think affect data quality. Mostly what you noticed in the Maternal Health data.
- For each problem, propose actions that could lead to improvements in data quality.
- You have 15 minutes to discuss in your subgroups before reporting back for session.

Spot the data quality issues

Below is one row from the ANC register. Take 3 minutes individually. Then we discuss as a group: how many distinct quality issues can you find, and which dimension does each belong to?

Field	Recorded value
client_name	NAKISIGE ROSE
client_no	189
ageyrs	88
anc_gravida	1
anc_parity	3
sbp / dbp	250 / 175
hb_levels_rslts	NIL
anc_diagnosis__2	ISS IN PREGNANCY
dent (visit date)	2027-02-15
expected_yrdlvry	23

YOUR TASK

- Find as many issues as you can in this row
- Match each to one of the six dimensions
- For each, suggest what action you would take

Discuss in pairs (3 min), then we surface findings as a group (5 min). Expected count: 7–9 distinct issues.

From assessment to action

An assessment is only useful if it changes what happens next. The audit produces three kinds of action.



FIX

In the data

Re-derive bad fields, drop impossible rows, impute where defensible, deduplicate via linkage.



FEED BACK

To the facility

Share completeness reports with clinic staff so workflows and forms can be improved upstream.



DOCUMENT

In the paper

Report quality metrics transparently. Every analytic decision rests on a quality finding.

END OF PART 1

Now let's use it.

⌘ *We should be here at minute 60*

The paper we are building

WORKING TITLE

Readiness of routine facilities for a digital perinatal cohort in rural Uganda: a data-quality assessment of antenatal records and their AI-readiness

PRIMARY AIM

To assess the completeness, consistency, and suitability of routine maternal health records at Mayuge HC IV and Buluba Hospital for supporting a digital perinatal cohort and artificial-intelligence applications.

SPECIFIC OBJECTIVES

1. Map available variables across ANC records
2. Assess completeness — variable and encounter level
3. Examine internal consistency and adherence to standards
4. Quantify missingness patterns and duplicates
5. Score AI-readiness across five domains

Our analysis plan in five steps

1

Load & inspect

Read the cleaned CSV,
parse dates and types,
build year.

2

Completeness

% missing per variable, by
facility and year.

3

Consistency

Logical and range rules;
count failures per rule.

4

Missingness

Heatmaps, upset plots,
test for patterned
missingness.

5

Readiness score

Composite score across
five domains.

Each step produces a table or figure that goes straight into the manuscript.

Step 1 — Load and inspect

```
library(tidyverse); library(nanjar); library(janitor); library(lubridate)

anc <- read_csv('ANC_final_clean.csv') %>% clean_names()

anc <- anc %>%
  mutate(
    anc_date = dmy(dent),          # ← VISIT date
    year      = year(anc_date),
    facility = case_when(
      str_detect(fac_num, regex('Mayuge', TRUE)) ~ 'Mayuge HC IV',
      str_detect(fac_num, regex('Buluba', TRUE)) ~ 'Buluba Hospital'
    )
  ) %>% filter(year %in% c(2023, 2024))

glimpse(anc); nrow(anc); table(anc$facility, anc$year)
```

WHAT WE GET

- n records loaded with the visit-date issue fixed
- Facility variable normalised to two values
- Counts of records per facility and year
- A clean object to feed into Steps 2-5

Step 2 — Variable-level completeness

```
# Variable-level completeness
miss_by_var <- anc %>%
  miss_var_summary() %>%
  arrange(desc(pct_miss))

miss_by_var %>% slice_head(n = 20)

# By facility
anc %>%
  group_by(facility) %>%
  miss_var_summary() %>%
  pivot_wider(names_from = facility,
              values_from = pct_miss) %>%
  arrange(desc(`Mayuge HC IV`))
```

OUTPUT

Table 1

Variable-level completeness, ranked from worst to best, split by facility. The top of this table tells us which fields cannot be used downstream until they are improved.

Step 3 — Consistency checks

```
checks <- anc %>% transmute(
  age_range      = ageyrs < 12 | ageyrs > 55,
  bp_range       = sbp < 60 | sbp > 220 |
                  dbp < 30 | dbp > 140,
  hb_range       = hb_levels_rslts < 4 |
                  hb_levels_rslts > 18,
  parity_logic   = anc_parity > anc_gravida,
  visit_order    = anavisit < 1 | anavisit > 12,
  gest_age       = anc_gestationalage < 4 |
                  anc_gestationalage > 44
)

checks %>% summarise(across(everything(),
                           ~mean(., na.rm = TRUE)))
```

OUTPUT

Table 2

Percentage of records failing each consistency rule. We expect single-digit failure rates; anything higher signals a workflow issue we should report.

Step 4 — Missingness patterns

```
# Variable-level missingness, faceted by facility
naniar::gg_miss_var(anc, facet = facility)

# Upset plot — which variables go missing together?
core <- c('ageyrs', 'anc_gravida', 'anc_parity',
          'sbp', 'dbp', 'hb_levels_rslts',
          'anc_bloodgroup', 'emtct_codes')
naniar::gg_miss_upset(anc %>% select(all_of(core)))

# Test patterned missingness for one key variable
anc %>% mutate(hb_missing = is.na(hb_levels_rslts)) %>%
  glm(hb_missing ~ facility + year + ageyrs +
      anc_gestationalage, data = .,
      family = binomial) %>%
  broom::tidy()
```

OUTPUTS

Figure 1 — Missingness bar plot

Figure 2 — Upset plot of co-missingness

Table 3 — Predictors of missing Hb

Step 5 — The AI-readiness score

We combine the previous steps into a single transparent score, 0-25, broken down by five domains. The rubric is published with the paper.

Domain	Definition	Max score
Completeness	% of core variables with low missingness	/ 5
Consistency	% of records passing all logical rules	/ 5
Linkability	Quality of identifiers for cross-record linkage	/ 5
Outcome availability	Whether the outcomes of interest are recorded	/ 5
Temporal structure	Date integrity and proper visit ordering	/ 5
Total AI-readiness score		/ 25

Run your first checks — together

Open RStudio. Load the ANC dataset. We will work through three tasks together. The facilitator runs the code; you follow on your machine and report what you see.

Task A5 min	Task B5 min	Task C5 min
Variable completeness WHAT Compute % missing for hb_levels_rslts, sbp, anc_iptdose, and emtct_codes. REPORT BACK Which is missing most often? Does it differ by facility?	One consistency rule WHAT Count rows where anc_parity > anc_gravida. REPORT BACK How many records fail this rule? What proportion of the dataset?	First completeness chart WHAT Make a horizontal bar plot of the top 15 most-missing variables. REPORT BACK Which three variables are most concerning, and why?

What we walk out of this session with

TABLES

T1 — Variable-level completeness, by facility

T2 — Consistency rule failure rates

T3 — Predictors of missingness for Hb

T4 — Adherence to WHO/MOH care standards

T5 — AI-readiness scorecard

FIGURES

F1 — Variable missingness bar plot

F2 — Upset plot of co-missingness

F3 — Per-encounter completeness density

F4 — Consistency violations by facility

F5 — Readiness radar by domain

Manuscript outline

Introduction

Routine data as a foundation for AI in maternal health; the LMIC gap; the study question.

Methods

Setting, dataset, six DQ dimensions, indicators against WHO/MOH, the readiness rubric.

Results

Completeness profile, missingness patterns, consistency failures, duplicates, adherence to standards, readiness scores by facility and year.

Discussion

Implications for cohort design and AI applications; system-level recommendations; limitations.

Conclusion

Where Mayuge and Buluba sit on the readiness spectrum; what must change before AI deployment.

Next steps after today

This month	Run full analysis end-to-end on the cleaned dataset	<i>Lead analyst</i>
Month 2	Draft Methods and Results sections from the outputs	<i>Lead author + analyst</i>
Month 3	Internal review of draft; quality of presentation	<i>All co-authors</i>
Month 4	Submit pre-print to OSF and circulate for feedback	<i>Lead author</i>
Month 5-6	Revise; format to target journal; submit	<i>Lead author + corresponding</i>

Three things to take home



Quality has six dimensions

Completeness, consistency, accuracy, timeliness, validity, uniqueness. A record can pass one and fail another.



Measurement, not opinion

Every dimension is measured with a number. 'The data looks fine' does not appear in the paper.



Assessment drives action

Findings feed back to the facility, into the data pipeline, and into the manuscript — in that order of importance.

END OF WORKSHOP — START OF THE PAPER

**We have the framework,
the data, and the plan.**

What we do between now and next week determines whether this becomes a published paper or a half-finished script. Let's keep momentum.

Questions, comments, who-does-what